

VLADISLAV SHUBIN

Master student

Stockholm, Sweden

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RESEARCH INTERESTS

- Spectral Theory, Quantum Graphs, Operator Theory, Mathematical Physics.
- Coding Theory, Combinatorics
- Combinatorial Number Theory, Algebraic Number Theory

EDUCATION

Stockholm University / KTH Royal Institute of Technology

Stockholm, Sweden

Master in Mathematics

Sep 2023 – Present

- **Courses:** Advanced Real Analysis, Type Theory, Algebraic Geometry
- **Activities:** Gave a seminar talk on formal philosophy.

Master Thesis: “Berry Phase for Quantum Graphs”

- Supervisor: **Pavel Kurasov**

Contributions:

- Studied *topology change* via parameter-dependent vertex conditions.
- Derived explicit continuous eigenfunction families.
- Showed that topology change induces *a nontrivial Berry phase* π for real-valued eigenfunctions.
- Established a connection between topology change and the structure of operator domains.

Petrozavodsk University

Petrozavodsk, Russia

Bachelor in Mathematics

Sep 2019 – Aug 2023

- **Courses:** Applied Statistics, Actuarial Mathematics, Finance Mathematics, Random Processes
- **Activities:** Worked as a teacher assistant and popular lecturer with Vladimir Kuznetsov.

Bachelor Thesis: “Bernstein Polynomials and MacWilliams Transform”

- Supervisor: **Nikita Gogin, Vladimir Kuznetsov**

Contributions:

- Studied MacWilliams (Krawtchouk) transform and its applications in approximation theory.
- Found new formulae for Bernstein and Chebyshev polynomials.
- Developed an algorithm for computing Bernstein polynomials.

TEACHING EXPERIENCE

Teacher Assistant

2021-2023

- **Activities:** Conducting exams, creating learning materials, and giving lectures in computer science and mathematics.

Private Teacher

2017-2021

- **Activities:** Private teaching on Programming in C/C++ and Python.

PREPRINTS

1. Pavel Kurasov, Vladislav Shubin, Axel Tibbling. “Berry’s phase under topology change”. doi: <https://doi.org/10.48550/arXiv.2605.10798>

CONFERENCE PUBLICATIONS

1. Vladislav Shubin. “Investigation of φ -radical numbers”. In: 73rd Scientific Conference of Students and Young Scientist. Petrozavodsk, 2021.
2. Vladislav Shubin and Nikita Gogin. “Bernstein Polynomials and MacWilliams transform”. In: Polynomial Computer Algebra, 2023.
3. Vladislav Shubin and Nikita Gogin. “Binomial Coefficients as Functions of their Denominator; Another Primality Criteria for Natural Integers”. Polynomial Computer Algebra, 2024.
4. Vladislav Shubin and Nikita Gogin. “Prime Power Conjecture for Projective Planes”. Polynomial Computer Algebra, 2025.
5. Vladislav Shubin and Ivan Ushakov. “MacWilliams Transform with Binomial Weights”. Polynomial Computer Algebra, 2026.

OEIS contributions

I authored the following OEIS sequences: A355045, A355059, A337775, A337776, and also contributed to A000108

SKILLS, LANGUAGES, INTERESTS

- **Math Skills:**
 - Quantum Graphs, Spectral Theory, Functional Analysis
 - Coding Theory, Combinatorics
- **Programming Skills:**
 - Typst & LaTeX
 - Lean
 - Python
 - C/C++ & Rust
- **Languages:** English (C1), Russian (native)
- **Interests:**
 - Classical guitar (7-year formal training)
 - Formal philosophy (coursework at Higher School of Economics)
 - Programming (C/C++, Linux systems)